Introduction to Computer Science Fall 2026
Homework 7

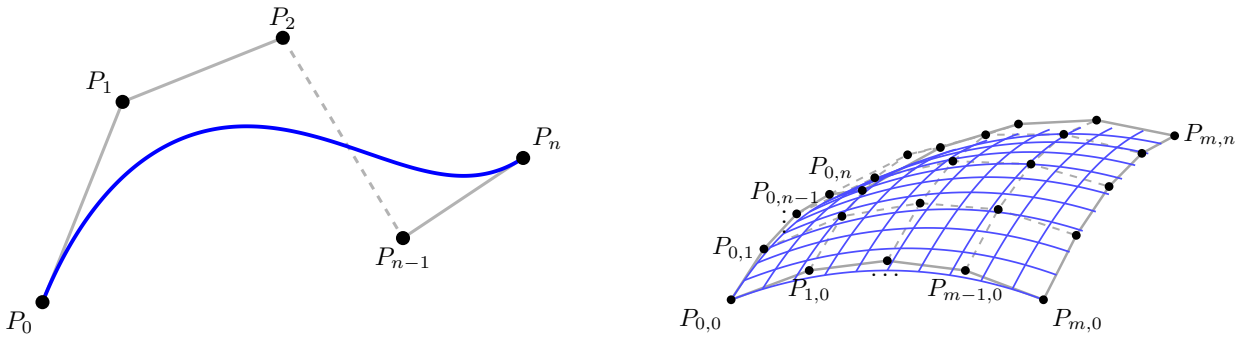
Due: November 15, 2026 at 23:58:59
No Copy-and-Paste. No Late Submission.

Instructions. Please write your answers to each problem using LaTeX and submit the compiled PDF file. If you are unfamiliar with LaTeX, you may refer to https://latex-tutorial.com/ or other tutorials for help.

If you find a problem difficult to understand or solve, you may use external resources, including search engines (such as Google), online references (such as Wikipedia), and AI tools (such as DeepSeek), to study relevant concepts and assist your learning. However, you are responsible for understanding your submitted solution, and **please make sure every word in your submission is written by yourself to avoid plagiarism**.

If you use any AI tool while working on the assignment, please include a short disclosure at the end of your submission stating which tool you used and briefly describing how you used it. Undisclosed use of AI tools may be subject to point deductions.

1 Bézier Curves and Surfaces (3 points)



(a) Bézier curve (b) Bézier surface

Figure 1: An example Bézier curve and surface.

(a) Bézier Curve (1 point)

Consider $n+1$ fixed control points $\mathbf{P}_0, \mathbf{P}_1, \ldots, \mathbf{P}_n$ in $\mathbb{R}^2$, as shown in Fig. 1(a).

A Bézier curve is constructed by repeatedly applying linear interpolation using the same parameter t. In the first round, interpolate each pair of neighboring control points to obtain n new points. In the second round, interpolate neighboring points among these new points to obtain $n-1$ points. Continue this process until only one point remains.

Note that although the original control points are fixed, all intermediate points depend on t and move as t varies. The final point, denoted by $\mathbf{B}(t)$, also moves with t. As t varies from 0 to 1, its trajectory forms the Bézier curve.

Derive an explicit expression for $\mathbf{B}(t)$ as a weighted sum of the original control points, i.e., $\mathbf{B}(t) = \sum_{i=0}^{n} w_i(t)\mathbf{P}_i$. Specify the weight $w_i(t)$ of each $\mathbf{P}_i$ in terms of n, i, and t.

(b) Bézier Surface (1 point)

Now consider a rectangular grid of fixed control points

$$\mathbf{P}_{ij} \in \mathbb{R}^3, \qquad i = 0, \ldots, m, \quad j = 0, \ldots, n,$$

as shown in Fig. 1(b).

A Bézier surface can be constructed by applying Bézier interpolation in two directions.

First, for each fixed j, use the control points $\mathbf{P}_{0j}, \ldots, \mathbf{P}_{mj}$ to construct a Bézier curve $\mathbf{Q}_j(u)$, where $u \in [0, 1]$. This produces $n + 1$ Bézier curves.

For any fixed u, take one point $\mathbf{Q}_j(u)$ from each of these curves. Treat these $n+1$ points as control points of another Bézier curve, and apply Bézier interpolation using a second parameter $v \in [0, 1]$. The resulting point is denoted by $\mathbf{S}(u, v)$.

As u and v vary independently over $[0, 1]$, the points $\mathbf{S}(u, v)$ form the Bézier surface.

Using your result from part (a), derive an explicit expression for $\mathbf{S}(u, v)$ as a weighted sum of the original control points $\mathbf{P}_{ij}$, i.e., $\mathbf{S}(u, v) = \sum_{i=0}^{m} \sum_{j=0}^{n} w_{ij}(u, v)\mathbf{P}_{ij}$. Specify the weight $w_{ij}(u, v)$ of each $\mathbf{P}_{ij}$ in terms of m, n, i, j, u, and v.

(c) Separability of Weights (1 point)

For the Bézier surface in part (b), can every weight $w_{ij}(u, v)$ be expressed as the product of a function of u and a function of v, i.e., $w_{ij}(u, v) = f_i(u)g_j(v)$? Justify your answer.

2 Recursive Ray Tracing (3 Points)

Ray tracing is a rendering technique that generates an image by tracing rays through a 3D scene. For each pixel, a *primary ray* is cast from the camera into the scene. When the ray intersects an object, the color of that pixel is determined by the material and lighting at the intersection.

For reflective or transparent materials, the color may also depend on light arriving from other directions. To account for this, the algorithm generates *secondary rays*, such as reflected or refracted rays, and traces them recursively. The colors returned by these rays are combined to compute the final pixel color (Fig. 2).

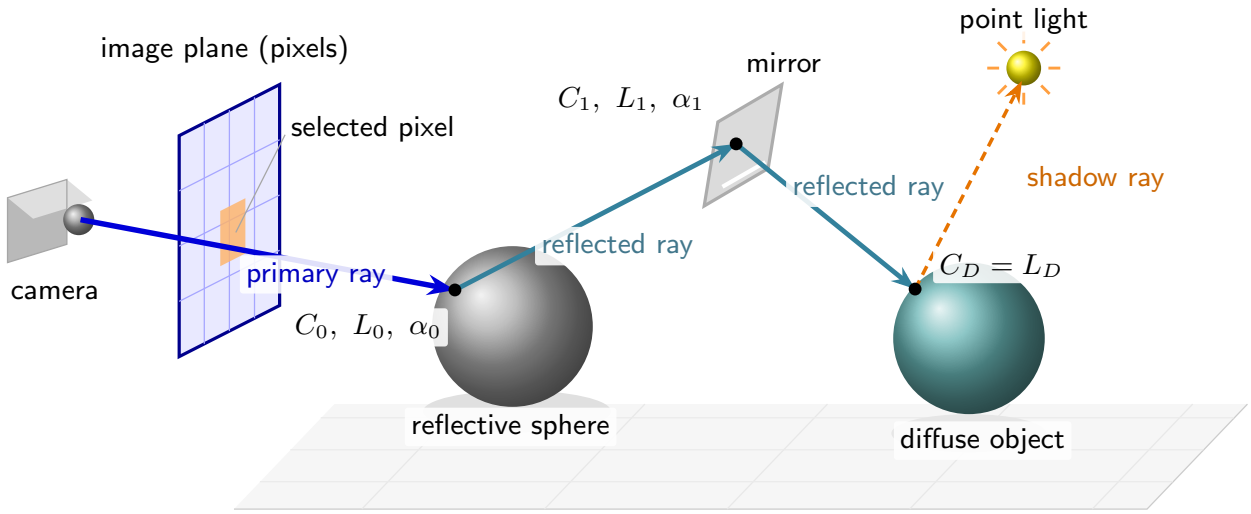


Figure 2: A 3D illustration of recursive ray tracing, where a primary ray passes through a pixel on the image plane and hits a reflective object.

(a) Color Computation (2 Points)

Consider a simplified ray tracing model that follows a single ray path. A primary ray is cast from the camera. Whenever it hits a reflective surface, one reflected ray is generated and traced further.

At each intersection, the returned color is computed as

$$C_i = (1 - \alpha_i)L_i + \alpha_i C_{i+1},$$

where L_i is the local RGB color, $\alpha_i \in [0, 1]$ is the reflection weight, and C_{i+1} is the color returned by the next reflected ray.

Suppose the ray undergoes two reflections, with the following values:

$$L_0 = (1, 0, 0), \quad \alpha_0 = \frac{1}{2},$$
$$L_1 = (0, 1, 0), \quad \alpha_1 = \frac{1}{4}.$$

The final ray hits a non-reflective surface with color $C_2 = (0, 0, 1)$.

Compute the returned colors C_1 and C_0. What is the final RGB color of the pixel?

(b) Number of Rays (2 Points)

In general, a ray may generate multiple secondary rays depending on the material it intersects. For example, a mirror generates a reflected ray, while a transparent surface may generate both reflected and refracted rays.

Consider a simplified probabilistic model. Whenever a ray hits an object, one of the following three cases occurs independently:

- With probability $1 - p - q$, the ray hits a diffuse surface and generates no secondary rays.
- With probability p, the ray hits a mirror and generates one reflected ray.
- With probability q, the ray hits a transparent surface and generates two secondary rays: one reflected and one refracted.

Here, $p, q \geq 0$ and $p + q \leq 1$. Assume that every generated ray intersects an object and follows the same probability distribution.

The primary ray starts at depth 0. Secondary rays are traced recursively until a maximum depth d is reached or a diffuse surface is hit, where no further rays are generated.

Derive the expected total number of rays $\mathbb{E}[N_{\text{total}}]$ traced for a single pixel, including all rays at depths 0 through d, in terms of p, q, and d.

3 Mass-Spring Simulation (4 Points)

Mass-spring systems are widely used in computer graphics to simulate deformable objects such as cloth and soft bodies. A key challenge in physical simulation is to approximate continuous motion using discrete time steps while maintaining numerical stability.

Consider the one-dimensional mass-spring system shown in Fig. 3. A particle of mass $m > 0$ is connected to a fixed wall by a spring with stiffness $k > 0$. Let x denote the displacement from equilibrium and $v = \dot{x}$ denote the velocity. Ignore gravity, damping, and collisions.

The elastic potential energy and total mechanical energy are defined as

$$U(x) = \frac{1}{2}kx^2, \qquad E(x, v) = \frac{1}{2}mv^2 + \frac{1}{2}kx^2.$$

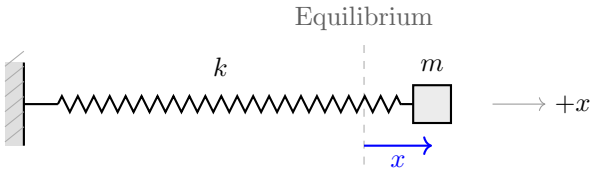


Figure 3: An one-dimensional mass-spring system used for this problem.

(a) Equation of Motion (1 Point)

The spring force is the negative derivative of the potential energy with respect to position: $f = -dU/dx$.

Using Newton's second law, the equation of motion can be written as

$$m\ddot{x} = cx.$$

Determine the coefficient c in terms of k.

(b) Explicit Euler (1 Point)

In a computer simulation, time is divided into discrete steps of size $h > 0$. Let x^t and v^t denote the position and velocity at time step t.

Explicit Euler uses the current velocity and acceleration to update the state:

$$x^{t+1} = x^t + hv^t, \qquad v^{t+1} = v^t + ha^t.$$

Using the equation of motion from part (a), derive the update formulas for x^{t+1} and v^{t+1}.

Then derive the change in mechanical energy $E^{t+1} - E^t$ in terms of E^t, h, k, and m.

(c) Implicit Euler (1 Point)

Implicit Euler instead uses the updated velocity and acceleration:

$$x^{t+1} = x^t + hv^{t+1}, \qquad v^{t+1} = v^t + ha^{t+1}.$$

Unlike Explicit Euler, the updated position and velocity depend on each other and must be solved together.

Using the equation of motion from part (a), derive explicit expressions for x^{t+1} and v^{t+1} in terms of x^t, v^t, h, k, and m.

Then derive the change in mechanical energy $E^{t+1} - E^t$ in terms of E^t, h, k, and m.

(d) Energy-Preserving Integration (1 Point)

Consider a hybrid integration method that combines Explicit Euler and Implicit Euler using a parameter $\theta \in [0, 1]$:

$$x^{t+1} = x^t + h\big((1-\theta)v^t + \theta v^{t+1}\big),$$
$$v^{t+1} = v^t + h\big((1-\theta)a^t + \theta a^{t+1}\big).$$

Here, $\theta = 0$ corresponds to Explicit Euler, while $\theta = 1$ corresponds to Implicit Euler.

Find a value of θ such that the hybrid method preserves the total mechanical energy for any initial state and any time step $h > 0$, i.e.,

$$E^{t+1} = E^t.$$

Prove that your choice satisfies this condition.